Lecture 5

5.1 The Fourier Integral Representation

5.2 The Fourier Transform

5.1 The Fourier Integral Representation

フーリエ積分表現

5.1 The Fourier Integral Representation

Review Lecture 2

Theorem 2.2 Fourier Series Representation: Arbitrary Period (任意周期)

Suppose that f is a $2p$-periodic piecewise **smooth** function. The Fourier series of f is given by

$$a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{p} x + b_n \sin \frac{n\pi}{p} x \right) \quad (2.7)$$

where

$$a_0 = \frac{1}{2p} \int_{-p}^{p} f(x)\, dx \quad (2.8)$$

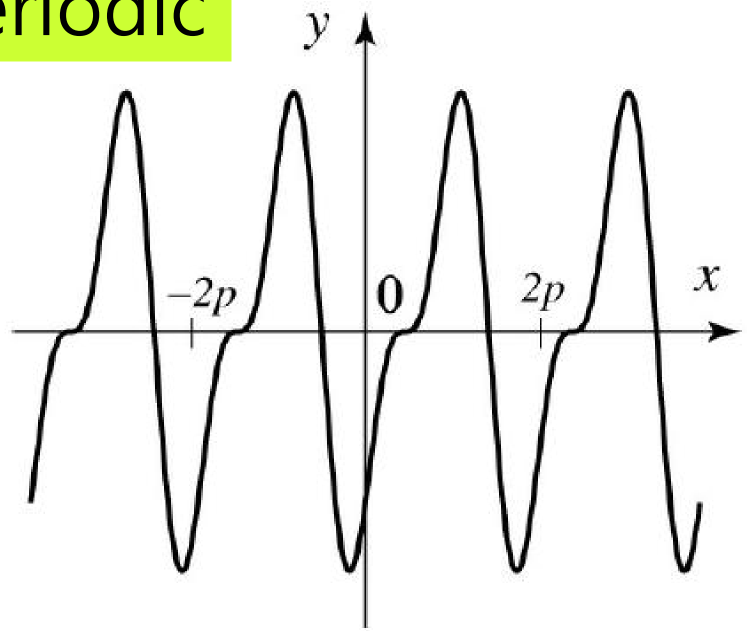
$$a_n = \frac{1}{p} \int_{-p}^{p} f(x) \cos \frac{n\pi}{p} x\, dx \quad (n = 1, 2, \ldots) \quad (2.9)$$

$$b_n = \frac{1}{p} \int_{-p}^{p} f(x) \sin \frac{n\pi}{p} x\, dx \quad (n = 1, 2, \ldots) \quad (2.10)$$

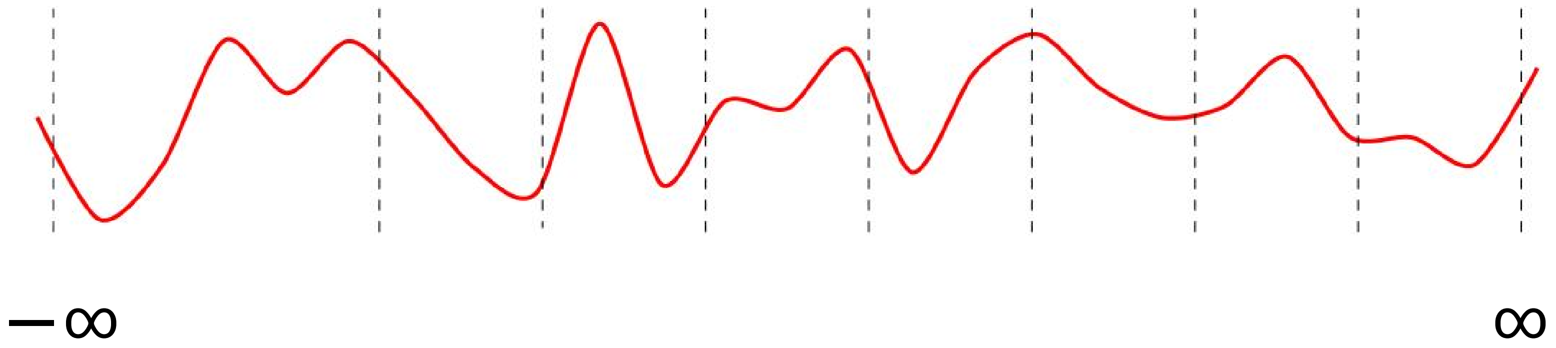
The Fourier series **converges to** $f(x)$ if f is **continuous at** x and to $\frac{f(x-) + f(x+)}{2}$ otherwise.

5.1 The Fourier Integral Representation

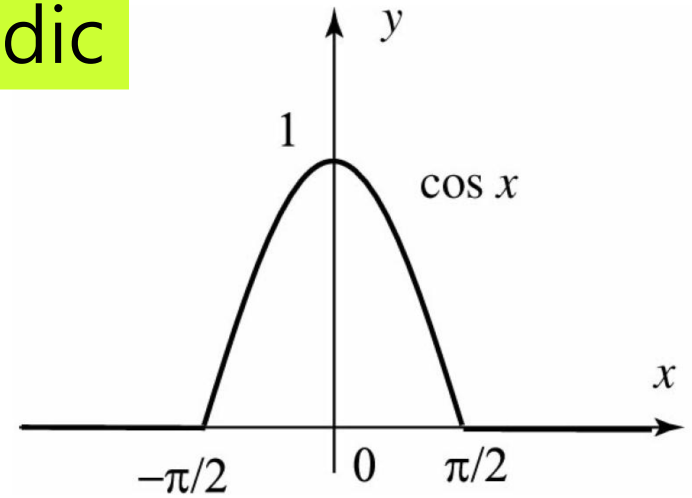
Periodic



A 2p-periodic function



Non-periodic



$$\begin{cases} \cos x & \text{if } |x| < \pi/2, \\ 0 & \text{if } |x| > \pi/2. \end{cases}$$

Now **suppose** that *f* **is defined on the entire real line but is NOT periodic**.

Q: Can we represent *f* by a Fourier series?

A: NO. But we can represent *f* as Fourier Integral.

5.1 The Fourier Integral Representation

Theorem 2.2 Fourier Series Representation: Arbitrary Period (任意周期)

Suppose that f is a $2p$-periodic piecewise **smooth** function. The Fourier series of f is given by

$$a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi}{p}x + b_n \sin\frac{n\pi}{p}x\right) \tag{2.7}$$

where

$$a_0 = \frac{1}{2p}\int_{-p}^{p} f(x)\,dx \tag{2.8}$$

$$a_n = \frac{1}{p}\int_{-p}^{p} f(x)\cos\frac{n\pi}{p}x\,dx \quad (n = 1, 2, \dots) \tag{2.9}$$

$$b_n = \frac{1}{p}\int_{-p}^{p} f(x)\sin\frac{n\pi}{p}x\,dx \quad (n = 1, 2, \dots) \tag{2.10}$$

The Fourier series converges to $f(x)$ if f is **continuous at** x and to $\frac{f(x-)+f(x+)}{2}$ otherwise.

When $p \to \infty$, what will happen to the theorem?

$$\lim_{p\to\infty} a_0 = \lim_{p\to\infty}\frac{1}{2p}\int_{-p}^{p} f_p(x)\,dx = 0$$

For a large p

$$a_n \approx \frac{1}{p}\int_{-\infty}^{\infty} f_p(x)\cos\frac{n\pi}{p}x\,dx \quad (n = 1, 2, \dots)$$

$$= \Delta\omega \frac{1}{\pi}\int_{-\infty}^{\infty} f_p(x)\cos\omega_n x\,dx \qquad \omega_n = \frac{n\pi}{p} \qquad \Delta\omega = \frac{\pi}{p}$$

$$= \Delta\omega \cdot A(\omega)$$

Similarly, $b_n \approx \Delta\omega \cdot B(\omega)$

$$\int_{-\infty}^{\infty}|f(x)|dx < \infty \quad \Rightarrow \quad f(x) = \lim_{p\to\infty} f_p(x) \approx 0 + \sum_{n=1}^{\infty}\left([\Delta\omega\cdot A(\omega)]\cos\omega_n x + [\Delta\omega\cdot B(\omega)]\sin\omega_n x\right)$$

$$= \sum_{n=1}^{\infty}\left(A(\omega)\cos\omega_n x + B(\omega)\sin\omega_n x\right)\Delta\omega$$

$n \to \infty \quad \omega_n \to \omega$

ω is nonnegative

$$\lim_{\Delta\omega\to 0} f(x) = \int_{0}^{\infty}[A(\omega)\cos\omega x + B(\omega)\sin\omega x]\,d\omega \quad (-\infty < x < \infty)$$

5.1 The Fourier Integral Representation

Theorem 5.1 Fourier Integral Representation

Suppose that f **is piecewise smooth on every finite interval** and that

$$\int_{-\infty}^{\infty} |f(x)|dx < \infty$$

Then f has the following **Fourier integral representation**

$$f(x) = \int_0^{\infty} [A(\omega)\cos\omega x + B(\omega)\sin\omega x]d\omega \quad (-\infty < x < \infty) \tag{5.1}$$

$$A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(t)\cos\omega t\, dt \tag{5.2}$$

$$B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(t)\sin\omega t\, dt \tag{5.3}$$

The integral in (5.1) converges to $f(x)$ if f is **continuous at** x and to $\dfrac{f(x+) + f(x-)}{2}$ otherwise.

5.1 The Fourier Integral Representation

Note **the similarity** between the Fourier series and Fourier integral

Fourier series

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi}{p}x + b_n \sin\frac{n\pi}{p}x\right) \quad (2.7)$$

$$a_0 = \frac{1}{2p}\int_{-p}^{p} f(x)\,dx$$

$$a_n = \frac{1}{p}\int_{-p}^{p} f(x)\cos\frac{n\pi}{p}x\,dx \quad (n = 1,\ 2,\ \dots)$$

$$b_n = \frac{1}{p}\int_{-p}^{p} f(x)\sin\frac{n\pi}{p}x\,dx \quad (n = 1,\ 2,\dots)$$

Fourier integral

$$f(x) = \int_0^{\infty}[A(\omega)\cos\omega x + B(\omega)\sin\omega x]d\omega \quad (-\infty < x < \infty) \quad (5.1)$$

$$A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(t)\cos\omega t\,dt \quad (5.2)$$

$$B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(t)\sin\omega t\,dt \quad (5.3)$$

① The **sum** in (2.7) is replaced by an **integral** in (5.1)

② The **integrals from** $-p$ **to** p are replaced by **integrals from** $-\infty$ **to** ∞ in (5.2) and (5.3).

③ In (5.2) and (5.3), the "Fourier coefficients" are computed **over a continuous range** $\omega \geq 0$, whereas the Fourier coefficients of a periodic function are computed **over a discrete range of values** $n = 0,\ 1,\ 2,\ \dots$.

5.1 The Fourier Integral Representation

Example 5.1 Find the Fourier integral representation of the function

$$f(x) = \begin{cases} 1 & if\ |x| \le 1 \\ 0 & otherwise \end{cases}$$

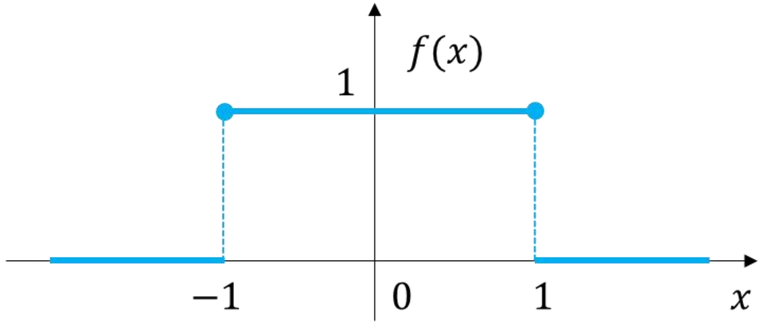


Figure 5.1 Figure of $f(x)$

Solution:

From (5.2),

$$A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(t)\cos\omega t\,dt$$
$$= \frac{1}{\pi}\int_{-1}^{1}\cos\omega t\,dt$$
$$= \left[\frac{\sin\omega t}{\pi\omega}\right]_{-1}^{1}$$
$$= \frac{1}{\pi}\,\frac{2\sin\omega}{\omega}$$

Since $f(x)$ is even, $f(x)\sin\omega x$ is odd, then by (2.4) in Lecture 2, $B(\omega)=0$.

If f is **odd**, then

$$\int_{-p}^{p} f(x)dx = 0 \qquad (2.4)$$

For $|x| \neq 1$ the function is continuous and Theorem 5.1 gives

$$f(x) = \int_0^\infty [A(\omega)\cos\omega x + B(\omega)\sin\omega x]\,d\omega \quad (-\infty < x < \infty)$$
$$= \frac{2}{\pi}\int_0^\infty \left[\frac{\sin\omega}{\omega}\cdot\cos\omega x + 0\right] d\omega$$
$$= \frac{2}{\pi}\int_0^\infty \frac{\sin\omega\cos\omega x}{\omega}\,d\omega$$

For $x = \pm 1$, which are the points of discontinuity of f, (5.1) of Theorem 5.1 yields the result $f(-1) = f(1) = \frac{1}{2}$.

Thus we have the Fourier integral representation of f

$$\frac{2}{\pi}\int_0^\infty \frac{\sin\omega\cos\omega x}{\omega}\,d\omega = \begin{cases} 1 & if\ |x|<1 \\ \frac{1}{2} & if\ |x|=1 \\ 0 & if\ |x|>1 \end{cases}$$

5.1 The Fourier Integral Representation

Setting $x = 0$ in the integral representation of Example 5.1 yields the important integral

$$\frac{2}{\pi}\int_0^\infty \frac{\sin\omega}{\omega}\,d\omega = 1$$

$$\int_0^\infty \frac{\sin\omega}{\omega}\,d\omega = \frac{\pi}{2} \qquad (5.4)$$

known as the **Dirichlet integral**, after the German mathematician Peter Gustave Dirichlet (1805–1859).

Peter Gustave Dirichlet
ペーター・グスタフ・ディリクレ
German mathematician
(1805–1859)

5.1 The Fourier Integral Representation

Example 5.2 Computing integrals via the Fourier integral

Let

$$f(x) = \begin{cases} \cos x & if\ |x| < \dfrac{\pi}{2} \\ 0 & if\ |x| > \dfrac{\pi}{2} \end{cases}$$

show that

$$\frac{2}{\pi}\int_0^\infty \frac{\cos\dfrac{\pi\omega}{2}}{1-\omega^2}\cos\omega x d\omega = \begin{cases} \cos x & if\ |x| < \dfrac{\pi}{2} \\ 0 & if\ |x| > \dfrac{\pi}{2} \end{cases}$$

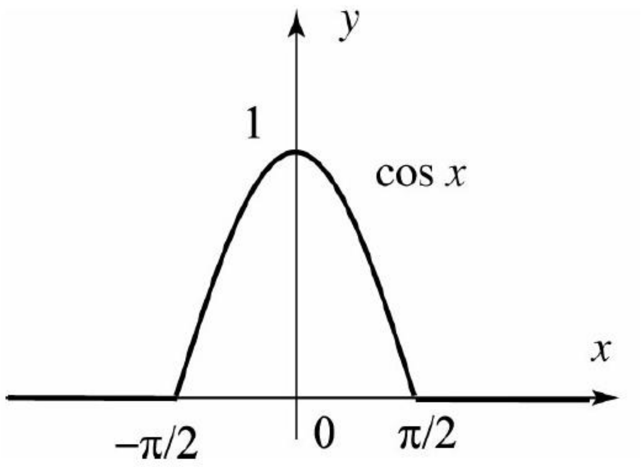


Figure 5.2 Figure of $f(x)$

5.1 The Fourier Integral Representation

Solution:

$f(x)$ is even and vanishes outside the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Thus $B(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(t)\sin wt\, dt = 0$ for all $w \geq 0$,

and

$$A(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(t)\cos wt\, dt$$

$$\Rightarrow A(w) = \frac{1}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \cos wx\, dx$$

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} \cos x \cos wx\, dx \quad \text{(by evenness)}$$

$$= \frac{1}{\pi}\int_0^{\frac{\pi}{2}} [\cos(x+wx) + \cos(x-wx)]\, dx \quad \text{(Trigonometric identity)}$$

$$= \frac{1}{\pi}\left[\left[\frac{\sin(1+w)x}{1+w}\right]_0^{\frac{\pi}{2}} + \left[\frac{\sin(1-w)x}{1-w}\right]_0^{\frac{\pi}{2}}\right] \quad (\text{for } w \neq 1)$$

$$= \frac{1}{\pi}\left[\frac{\sin[(1+w)\frac{\pi}{2}]}{1+w} - 0 + \frac{\sin[(1-w)\frac{\pi}{2}]}{1-w} - 0\right]$$

$$= \frac{1}{\pi}\left[\frac{\cos(\frac{w\pi}{2})}{1+w} + \frac{\cos(\frac{w\pi}{2})}{1-w}\right]$$

$$= \frac{2\cos(\frac{w\pi}{2})}{\pi(1-w^2)}$$

The case $w=1$ should be treated separately. We see $A(1) = \frac{1}{2}$. You can check that $\lim_{w\to 1} A(w) = A(1) = \frac{1}{2}$. Now using (5.1) in Theorem 5.1, we get

$$f(x) = \frac{2}{\pi}\int_0^{\infty} \frac{\cos(w\pi/2)}{1-w^2}\cos wx\, dw$$

Replacing $f(x)$ by its formula, we get the desired identity.

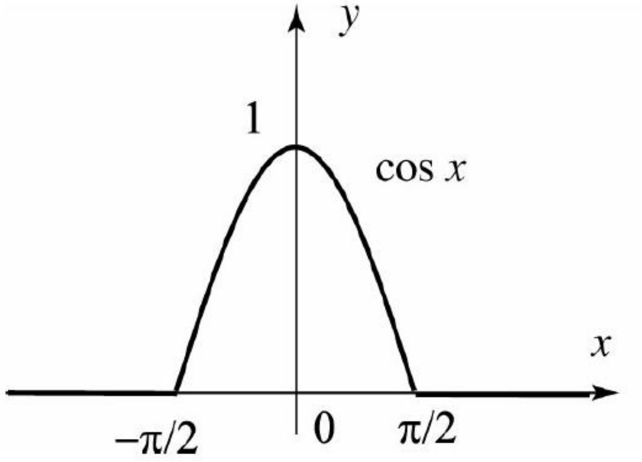


Figure 5.2 Figure of $f(x)$

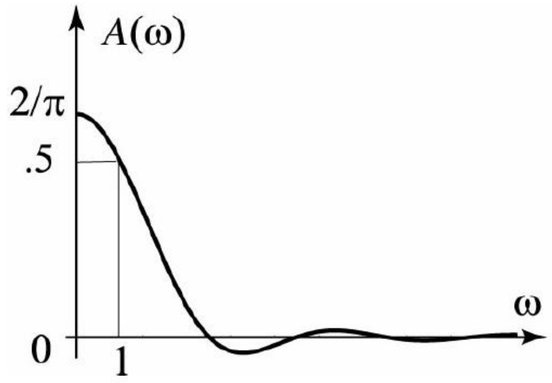


Figure 5.3 $A(\omega)$ is continuous.

5.1 The Fourier Integral Representation

*Partial Fourier Integrals and the Gibbs Phenomenon

In analogy with the partial sums of Fourier series, we define the partial Fourier integral of f by

$$S_\nu(x) = \int_0^\nu [A(\omega)\cos\omega x + B(\omega)\sin\omega x]\,d\omega \quad (\text{for } \nu > 0)$$

where $A(\omega)$ and $B(\omega)$ are given by (5.2) and (5.3). With this notation, Theorem 5.1 states

$$\lim_{\nu\to\infty} S_\nu(x) = \frac{f(x+) + f(x-)}{2}$$

Like Fourier series, near a point of discontinuity the Fourier integral exhibits a **Gibbs phenomenon**. To illustrate this phenomenon, we introduce the **sine integral function**

$$\mathrm{Si}(x) = \int_0^\infty \frac{\sin t}{t}\,dt \qquad (-\infty < x < \infty)$$

5.1 The Fourier Integral Representation

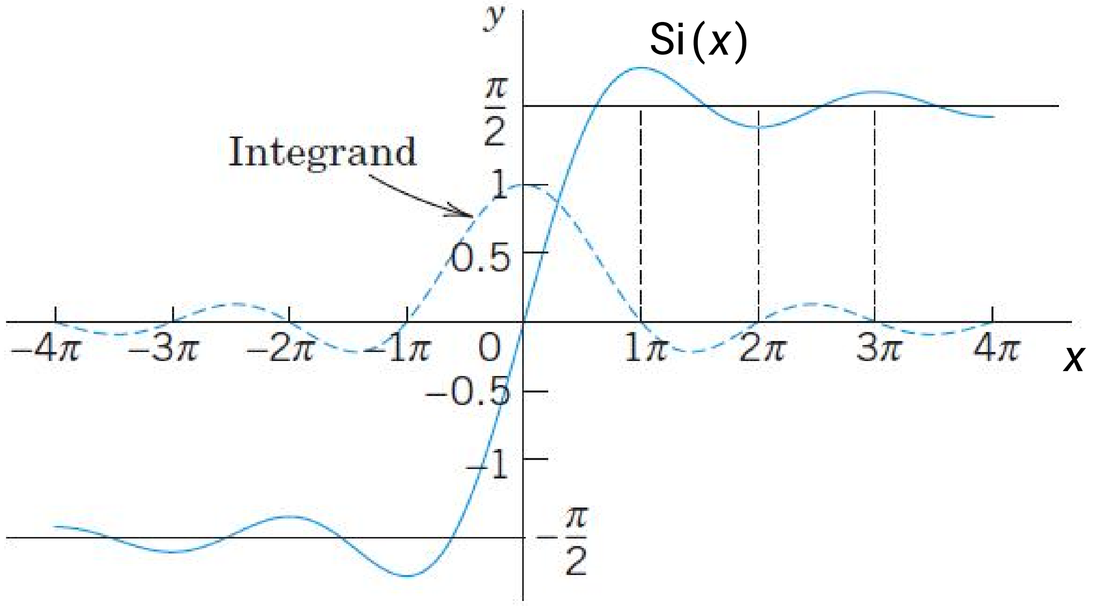


Figure 5.4 Sine integral $\mathrm{Si}(x)$ and integrand

Because of its frequent occurrence, the function $Si(x)$ is tabulated and is available as a standard function in most computer systems. See Figure 5.4 for its graph. From (5.4), it follows that

$$\lim_{x\to\infty} \mathrm{Si}(x) = \int_0^\infty \frac{\sin t}{t} dt = \frac{\pi}{2}$$

5.2 The Fourier Transform

フーリエ変換

5.2 The Fourier Transform

Consider a continuous piecewise smooth integrable function f.

Starting with the Fourier integral representation in Theorem 5.1,

we have

$$f(x) = \int_0^\infty [A(w)\cos wx + B(w)\sin wx]\, dw \quad (-\infty < x < \infty)$$

$$= \int_0^\infty \left[\frac{1}{\pi}\int_{-\infty}^\infty f(t)\cos wt\, dt \cdot \cos wx + \frac{1}{\pi}\int_{-\infty}^\infty f(t)\sin wt\, dt \cdot \sin wx\right] dw$$

$$= \frac{1}{\pi}\int_0^\infty \int_{-\infty}^\infty f(t)\,(\cos wt \cos wx + \sin wt \sin wx)\, dt\, dw$$

$$= \frac{1}{\pi}\int_0^\infty \int_{-\infty}^\infty f(t)\cos w(x-t)\, dt\, dw \qquad (\text{Because } \cos(a-b) = \cos a \cos b + \sin a \sin b)$$

$$= \frac{1}{\pi}\int_0^\infty \int_{-\infty}^\infty f(t)\left[\frac{1}{2}\left(e^{iw(x-t)} + e^{-iw(x-t)}\right)\right] dt\, dw$$

$$= \frac{1}{2\pi}\int_0^\infty \int_{-\infty}^\infty f(t)\, e^{iw(x-t)}\, dt\, dw + \frac{1}{2\pi}\int_0^\infty \int_{-\infty}^\infty f(t)\, e^{-iw(x-t)}\, dt\, dw$$

5.2 The Fourier Transform

If we change ω to $-\omega$ in the second term and adjust the limits on ω from $-\infty$ to 0, we obtain, after adding the two integrals,

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t)\, e^{i\omega(x-t)}\, dt\, d\omega$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\omega x} \underbrace{\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t)\, e^{-i\omega t}\, dt}_{\hat{f}(\omega)}\, d\omega$$

$\hat{f}(\omega) \Leftarrow$ our class

$$\left(= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega x} \underbrace{\int_{-\infty}^{\infty} f(t)\, e^{-i\omega t}\, dt}_{\hat{f}(\omega)}\, d\omega \right.$$

$\hat{f}(\omega) \Leftarrow$ other textbook $\Big)$

5.2 The Fourier Transform

This is the complex form of the Fourier integral representation, which features the following transform pair:

Script $\mathscr{F}$: Fourier Transform

Fourier Transform フーリエ変換 $\hat{f}(\omega) = \mathscr{F}[f(x)]$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x}\, dx \qquad (-\infty < \omega < \infty) \qquad (5.5)$$

Inverse Fourier Transform フーリエ逆変換 $f(x) = \mathscr{F}^{-1}[\hat{f}(\omega)]$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\omega x}\, \hat{f}(\omega) d\omega \qquad (-\infty < x < \infty) \qquad (5.6)$$

5.2 The Fourier Transform

Putting $\omega = 0$ in (5.5), we find that

$$\hat{f}(0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\, dx$$

Thus the value of the Fourier transform at $\omega = 0$ is equal to the signed area between the graph of $f(x)$ and the x-axis, multiplied by a factor of $1/\sqrt{2\pi}$.

5.2 The Fourier Transform

Example 5.3 **A Fourier transform**

(a) Find the **Fourier transform** of the function in Figure 5.5, given by

$$f(x) = \begin{cases} 1 & if\ |x| < a \\ 0 & if\ |x| > a \end{cases}$$

What is $\hat{f}(0)$?

(b) Express f as an **inverse Fourier transform**.

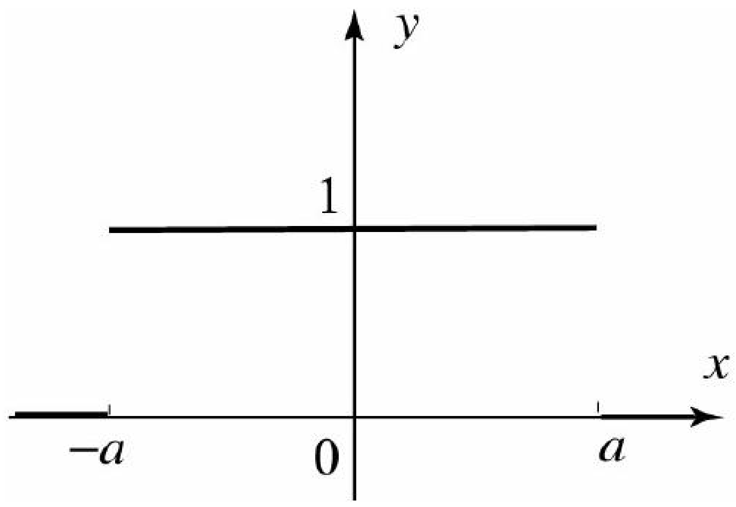


Figure 5.5 Figure of $f(x)$

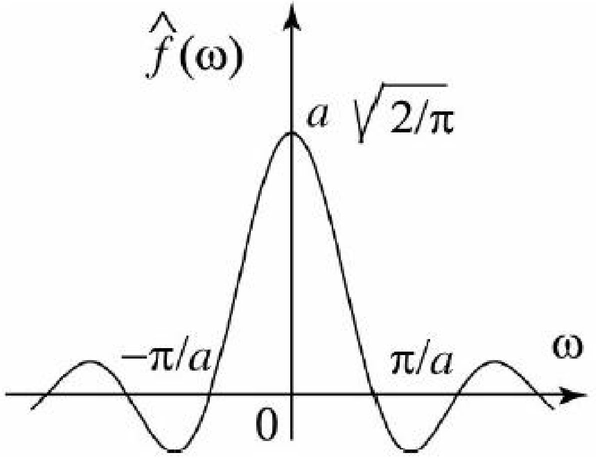


Figure 5.6 Figure of $\hat{f}$

5.2 The Fourier Transform

Solution:

(a) For $w \neq 0$ we have

$$\hat{f}(w) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} 1 \cdot e^{-iwx} dx$$

$$= \left[-\frac{1}{\sqrt{2\pi}\, iw} e^{-iwx} \right]_{-a}^{a} = \sqrt{\frac{2}{\pi}} \frac{\sin aw}{w}$$

For $w = 0$ we have $\lim_{w \to 0} \hat{f}(w) = \lim_{w \to 0} \sqrt{\frac{2}{\pi}} \frac{\sin aw}{w} = a\sqrt{\frac{2}{\pi}} = \hat{f}(0)$

it follows that $\hat{f}(w)$ is continuous at 0 (See Figure 5.6),

and we write

$$\hat{f}(w) = \sqrt{\frac{2}{\pi}} \frac{\sin aw}{w} \quad \text{for all } w$$

(b) To express f as an inverse Fourier transform,

we use (5.6) and set

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{iwx} \sqrt{\frac{2}{\pi}} \frac{\sin aw}{w} dw$$

$$= \frac{1}{\pi} \int_{-\infty}^{\infty} e^{iwx} \frac{\sin aw}{w} dw$$

$$= \frac{1}{\pi} \int_{-\infty}^{\infty} (\cos wx + i \sin wx) \frac{\sin aw}{w} dw$$

$$= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\cos wx \sin aw}{w} dw$$

because $\sin wx \frac{\sin aw}{w}$ is an odd function of w

and so its integral is zero.

When $a = 1$, this representation coincides with

the integral representation in Example 5.1.

The Fourier transform in Example 5.3 is continuous on the entire real line even though the function has jump discontinuities at $x = \pm a$.

In fact, it can be shown that **the Fourier transform of an integrable function is always continuous.**

5.2 The Fourier Transform

Example 5.4 **Computing Fourier transforms**

Find the **Fourier transform** of the function in Figure 5.7,

$$f(x) = \begin{cases} e^{-x} & if\ x > 0 \\ 0 & if\ x \leq 0 \end{cases}$$

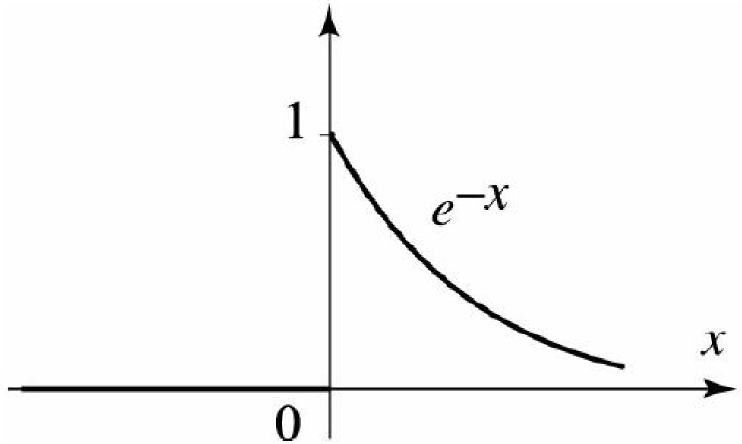


Figure 5.7 Figure of $f(x)$

Solution:

From (5.5), we have

$$\hat{f}(w) = \frac{1}{\sqrt{2\pi}} \int_0^\infty e^{-x} e^{-iwx}\, dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^\infty e^{-x(1+iw)}\, dx$$

$$= \left[-\frac{1}{\sqrt{2\pi}(1+iw)} e^{-iwx} e^{-x} \right]_0^\infty$$

Let's recall that if $z = a+ib$, then $|z| = \sqrt{a^2+b^2}$.

In particular, if $z = e^{-iwx}$, then

$$|e^{-iwx}| = |\cos(wx) - i\sin(wx)| = \sqrt{\cos^2(wx) + \sin^2(wx)} = 1$$

It follows that $\lim_{x\to\infty} |e^{-x(1+iw)}| = \lim_{x\to\infty} e^{-x} = 0$,

and so

$$\hat{f}(w) = \frac{1}{\sqrt{2\pi}(1+iw)} = \frac{1-iw}{\sqrt{2\pi}(1+w^2)}$$

Figure 5.8 shows the graph of the absolute value of $\hat{f}$. Here, it is worth noting that $\hat{f}$ and $|\hat{f}|$ are both continuous even though f is not.

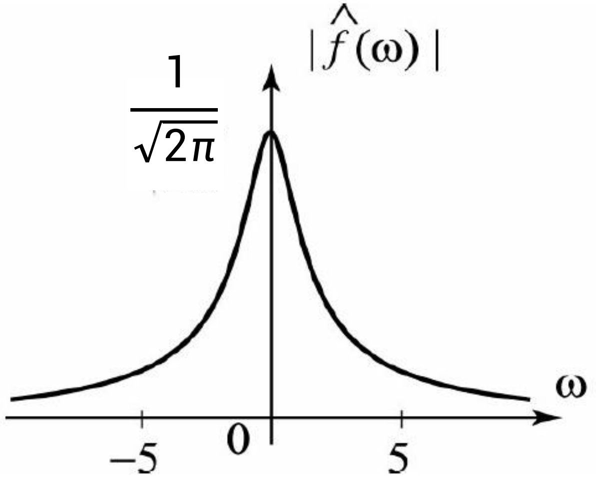


Figure 5.8 Figure of $|\hat{f}|$

Operational Properties

We shall **investigate the behavior of the Fourier transform** in connection with the **common operations on functions**:

- **linear combination (線型結合)**
- translation
- dilation
- differentiation
- multiplication by polynomials
- **convolution (畳み込み)**

Theorem 5.2 Linearity of Fourier Transform

The Fourier transform is **a linear operation**; that is, for **any integrable functions** f and g and **any real numbers** a and b,

$$\mathfrak{F}(af + bg) = a\mathfrak{F}(f) + b\mathfrak{F}(g) \qquad (5.7)$$

Proof:

From (5.5),

$$\mathcal{F}[af(x)+bg(x)] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}[af(x)+bg(x)]e^{-iwx}dx$$

$$= a\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}f(x)e^{-iwx}dx + b\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}g(x)e^{-iwx}dx$$

$$= a\mathcal{F}[f(x)] + b\mathcal{F}[g(x)]$$

5.2 The Fourier Transform

Theorem 5.3 Fourier Transforms of Derivatives (導函数)

(i) Suppose $f(x)$ is **piecewise smooth**, $f(x)$ and $f'(x)$ are **integrable**, and $f(x) \to 0$ as $|x| \to \infty$, then

$$\mathcal{F}(f') = i\omega\mathcal{F}(f) \tag{5.8}$$

(ii) If in addition $f''(x)$ is **integrable**, and $f'(x)$ is piecewise smooth and $f'(x) \to 0$ as $|x| \to \infty$, then

$$\mathcal{F}(f'') = i\omega\mathcal{F}(f') = -\omega^2\mathcal{F}(f) \tag{5.9}$$

(iii) In general, if f and $f^{(n)}(x)$ $(n = 1, 2, \ldots, N-1)$ are piecewise smooth and tend to 0 as $|x| \to \infty$,

and f and its derivatives of order up to n are integrable, then

$$\mathcal{F}(f^{(n)}) = (i\omega)^n\mathcal{F}(f) \tag{5.10}$$

5.2 The Fourier Transform

Proof:

(i) From the definition of the Fourier transform we have

$$\mathcal{F}[f'(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f'(x) e^{-iwx} dx$$

Integrating by parts, we obtain

$$\mathcal{F}[f'(x)] = \frac{1}{\sqrt{2\pi}} \left[f(x) e^{-iwx} \Big|_{-\infty}^{\infty} - (-iw) \int_{-\infty}^{\infty} f(x) e^{-iwx} dx \right]$$

Since $f(x) \to 0$ as $|x| \to \infty$, the desired result follows, namely

$$\mathcal{F}[f'(x)] = 0 + iw \mathcal{F}[f(x)]$$

$$= iw \mathcal{F}[f(x)]$$

(ii) A successive application of (5.8) is

$$\mathcal{F}[f''] = iw \mathcal{F}[f'] = (iw)^2 \mathcal{F}[f] = -w^2 \mathcal{F}[f]$$

(iii) Similarly for higher derivatives.

Example 5.5 Find the Fourier transform of xe^{-x^2}

Solution:

$$\mathcal{F}[xe^{-x^2}] = \mathcal{F}[-\frac{1}{2}(e^{-x^2})']$$

$$= -\frac{1}{2}\mathcal{F}[(e^{-x^2})']$$

$$= -\frac{1}{2} i w \mathcal{F}[e^{-x^2}]$$

$$= -\frac{1}{2} i w \frac{1}{\sqrt{2}} e^{-\frac{w^2}{4}}$$

$$= -\frac{iw}{2\sqrt{2}} e^{-\frac{w^2}{4}}$$

5.2 The Fourier Transform

Theorem 5.4 Derivatives of Fourier Transforms

(i) Suppose $f(x)$ and $xf(x)$ are integrable; then

$$\mathcal{F}\big(xf(x)\big)(\omega) = i[\hat{f}]'(\omega) = i\frac{d}{d\omega}\mathcal{F}(f)(\omega)$$

(ii) In general, if $f(x)$ and $x^n f(x)$ are integrable, then

$$\mathcal{F}\big(x^n f(x)\big) = i^n[\hat{f}]^{(n)}(\omega)$$

5.2 The Fourier Transform

Sketch of the proof:

To motivate (i), we assume that we can differentiate under the integral. Then

$$[\hat{f}]'(w) = \frac{d}{dw}\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x)e^{-iwx}dx$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x)\frac{d}{dw}e^{-iwx}dx$$

$$= -\frac{i}{\sqrt{2\pi}}\int_{-\infty}^{\infty} xf(x)e^{-iwx}dx$$

$$= -i\,\mathcal{F}[xf(x)](w).$$

and (i) follows upon multiplying both sides by i.

This proof is valid if for example f is smooth and vanishes outside a finite interval.

For an arbitrary function f, we can approximate f by functions that are smooth and vanish outside a finite interval.

The details are beyond the level of this class and will be omitted.

Part (ii) follows from (i).

Table of Fourier Transforms

$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \widehat{f}(\omega) e^{ix\omega}\, d\omega$	$\widehat{f}(\omega) = \mathcal{F}(f)(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x}\, dx$
1. $\begin{cases} 1 & \text{if } \lvert x\rvert < a \\ 0 & \text{if } \lvert x\rvert > a \end{cases}$	$\sqrt{\frac{2}{\pi}} \frac{\sin a\omega}{\omega}$
2. $\begin{cases} 1 & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$	$\frac{i\,(e^{-i\,b\omega} - e^{i\,a\omega})}{\sqrt{2\pi}\omega}$
3. $\begin{cases} 1 - \frac{\lvert x\rvert}{a} & \text{if } \lvert x\rvert < a \\ 0 & \text{if } \lvert x\rvert > a \end{cases} \quad a > 0$	$2\sqrt{\frac{2}{\pi}} \frac{\sin^2(\frac{a\,\omega}{2})}{a\,\omega^2}$
4. $\begin{cases} x & \text{if } \lvert x\rvert < a \\ 0 & \text{if } \lvert x\rvert > a \end{cases} \quad a > 0$	$i\sqrt{\frac{2}{\pi}} \frac{a\omega\cos(a\omega) - \sin(a\omega)}{\omega^2}$
5. $\begin{cases} \sin x & \text{if } \lvert x\rvert < \pi \\ 0 & \text{if } \lvert x\rvert > \pi \end{cases}$	$i\sqrt{\frac{2}{\pi}} \frac{\sin(\pi\omega)}{\omega^2 - 1}$
6. $\begin{cases} \sin(ax) & \text{if } \lvert x\rvert < b \\ 0 & \text{if } \lvert x\rvert > b \end{cases} \quad a, b > 0$	$i\sqrt{\frac{2}{\pi}} \frac{\omega\cos(b\omega)\sin(ab) - a\cos(ab)\sin(b\omega)}{\omega^2 - a^2}$
7. $\frac{1}{a^2 + x^2},\ a > 0$	$\sqrt{\frac{\pi}{2}} \frac{e^{-a\lvert\omega\rvert}}{a}$
8. $\frac{x}{a^2 + x^2},\ a > 0$	$-i\sqrt{\frac{\pi}{2}}\, \operatorname{sgn}\omega\, e^{-a\lvert\omega\rvert}$
9. $\sqrt{\frac{2}{\pi}} \frac{a}{1 + a^2x^2},\ a > 0$	$e^{-\frac{\lvert\omega\rvert}{a}}$
10. $\frac{\sin ax}{x},\ a > 0$	$\begin{cases} \sqrt{\frac{\pi}{2}} & \text{if } \lvert\omega\rvert < a \\ \frac{1}{2}\sqrt{\frac{\pi}{2}} & \text{if } \lvert\omega\rvert = a \\ 0 & \text{if } \lvert\omega\rvert > a \end{cases}$
11. $\frac{4}{\sqrt{2\pi}} \frac{\sin^2(\frac{1}{2}ax)}{ax^2},\ a > 0$	$\begin{cases} 1 - \frac{\lvert\omega\rvert}{a} & \text{if } \lvert\omega\rvert < a \\ 0 & \text{if } \lvert\omega\rvert > a \end{cases}$
12. $\frac{4}{\sqrt{2\pi}} \frac{\sin^2(ax) - \sin^2(\frac{1}{2}ax)}{ax^2},\ a > 0$	$\begin{cases} 1 & \text{if } \lvert x\rvert < a \\ (-x + 2a)/a & \text{if } a < x < 2a \\ (x + 2a)/a & \text{if } a < x < 2a \\ 0 & \text{if } \lvert x\rvert > 2a \end{cases}$
13. $e^{-a\lvert x\rvert},\ a > 0$	$\sqrt{\frac{2}{\pi}} \frac{a}{a^2 + \omega^2}$
14. $\begin{cases} e^{-ax} & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases},\ a > 0$	$\frac{1}{\sqrt{2\pi}} \frac{1}{a + i\omega}$
15. $\begin{cases} 0 & \text{if } x > 0 \\ e^{ax} & \text{if } x < 0 \end{cases},\ a > 0$	$\frac{1}{\sqrt{2\pi}} \frac{1}{a - i\omega}$
16. $\lvert x\rvert^n e^{-a\lvert x\rvert},\ a > 0,\ n > 0$	$\frac{\Gamma(n+1)}{\sqrt{2\pi}} \left(\frac{1}{(a - i\omega)^{1+n}} + \frac{1}{(a + i\omega)^{1+n}} \right)$

$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \widehat{f}(\omega) e^{ix\omega}\, d\omega$	$\widehat{f}(\omega) = \mathcal{F}(f)(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x}\, dx$
17. $e^{-\frac{a}{2}x^2},\ a > 0$	$\frac{1}{\sqrt{a}} e^{-\frac{\omega^2}{2a}}$
18. $e^{-ax^2},\ a > 0$	$\frac{1}{\sqrt{2a}} e^{-\frac{\omega^2}{4a}}$
19. $xe^{-\frac{a}{2}x^2},\ a > 0$	$\frac{-i\omega}{a^{3/2}} e^{-\frac{\omega^2}{2a}}$
20. $x^2 e^{-\frac{a}{2}x^2},\ a > 0$	$\frac{a - \omega^2}{a^{5/2}} e^{-\frac{\omega^2}{2a}}$
21. $x^3 e^{-\frac{a}{2}x^2},\ a > 0$	$\frac{-i\omega(3a - \omega^2)}{a^{7/2}} e^{-\frac{\omega^2}{2a}}$
22. $e^{-\frac{x^2}{2}} H_n(x)$, H_n, nth Hermite polynomial	$(-1)^n i^n e^{\frac{-\omega^2}{2}} H_n(\omega)$
23. $J_0(x)$, Bessel function of order 0	$\begin{cases} \sqrt{\frac{2}{\pi}} \frac{1}{\sqrt{1-\omega^2}} & \text{if } \lvert\omega\rvert < 1 \\ 0 & \text{if } \lvert\omega\rvert > 1 \end{cases}$
24. $J_n(x)$, Bessel function of order $n \geq 0$	$\begin{cases} \sqrt{\frac{2}{\pi}} \frac{(-i)^n}{\sqrt{1-\omega^2}} T_n(\omega) & \text{if } \lvert\omega\rvert < 1 \\ 0 & \text{if } \lvert\omega\rvert > 1 \end{cases}$ T_n, Chebyshev polynomial of degree n.

Special Transforms

25. $\mathcal{F}(\delta_0(x))(\omega) = \frac{1}{\sqrt{2\pi}}$

26 $\mathcal{F}(\delta_0(x - a))(\omega) = \frac{1}{\sqrt{2\pi}} e^{-ia\omega}$

27. $\mathcal{F}(\sqrt{\frac{2}{\pi}} \frac{1}{x})(\omega) = -i \operatorname{sgn}\omega$

28. $\mathcal{F}(e^{iax})(\omega) = \sqrt{2\pi}\delta_0(\omega - a)$

Operational Properties

29. $\mathcal{F}(af + bg)(\omega) = a\mathcal{F}(f) + b\mathcal{F}(g)$

30. $\mathcal{F}(f')(\omega) = i\omega\mathcal{F}(f)(\omega)$

31. $\mathcal{F}(f'')(\omega) = -\omega^2\mathcal{F}(f)(\omega)$

32. $\mathcal{F}(f^{(n)})(\omega) = (i\omega)^n\mathcal{F}(f)(\omega)$

33. $\mathcal{F}(xf(x))(\omega) = i\frac{d}{d\omega}\mathcal{F}(f)(\omega)$

34. $\mathcal{F}(x^n f(x))(\omega) = i^n \frac{d^n}{d\omega^n}\mathcal{F}(f)(\omega)$

35. $\mathcal{F}(f * g)(\omega) = \mathcal{F}(f)(\omega)\mathcal{F}(g)(\omega)$

36. $\mathcal{F}(f\,g)(\omega) = \mathcal{F}(f) * \mathcal{F}(g)(\omega)$

37. $\mathcal{F}(f(x - a))(\omega) = e^{-ia\omega}\mathcal{F}(f)(\omega)$

38. $\mathcal{F}(e^{iax}f(x))(\omega) = \mathcal{F}(f)(\omega - a)$

39. $\mathcal{F}(\cos(ax)f(x))(\omega) = \frac{\mathcal{F}(f)(\omega-a)+\mathcal{F}(f)(\omega+a)}{2}$

40. $\mathcal{F}(\sin(ax)f(x))(\omega) = \frac{\mathcal{F}(f)(\omega-a)-\mathcal{F}(f)(\omega+a)}{2i}$

41. $\mathcal{F}(f(ax))(\omega) = \frac{1}{\lvert a\rvert}\mathcal{F}(f)(\frac{\omega}{a}),\ a \neq 0$

42. $f(x) = \mathcal{F}(\widehat{f})(-x),\ \mathcal{F}(\mathcal{F}(f)) = f(-x)$

Review for Lecture 5

- Fourier Integral
- Fourier Transform and Inverse Fourier Transform
- Linearity of Fourier Transform
- Derivatives of Fourier Series
- Table of Fourier Transforms

Assignment

Please Check https://web-ext.u-aizu.ac.jp/~xiangli/teaching/MA05/index.html

Reading materials: Lecture 5 Slides, Section 7.1, 7.2 and Appendix B of the Textbook

References

[1] Nakhlé H. Asmar, *Partial Differential Equations with Fourier Series and Boundary Value Problems 2nd Edition*, 2004

[2] Wikipedia